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Fixation devices and prostheses for soft tissue connection to the same

Abstract

An orthopedic assembly includes a tibial prosthesis that includes a body that defines an anterior side and a posterior side. The body further incudes a recess in the anterior side of the joint prosthesis and a plurality of openings that extend through the body from the anterior side to the posterior side thereof. At least a first and second opening of the openings are positioned at respective lateral and medial sides of a longitudinal axis of the tibial prosthesis. A modular insert is positioned within the recess of the body such that at least a portion of the modular insert is positioned between the openings of the body. The modular insert is formed separately from the tibial prosthesis and has a porous outer surface to promote tissue ingrowth.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/545,020, filed on Dec. 8, 2021, which claims priority from U.S. application Ser. No. 16/524,534, filed on Jul. 29, 2019, which claims the benefit of the filing date of U.S. Provisional Application No. 62/712,491, filed on Jul. 31, 2018, and U.S. Provisional Application No. 62/791,461, filed on Jan. 11, 2019, the disclosures of which are hereby incorporated by reference in their entireties.

BACKGROUND OF THE INVENTION
(1) Joint replacement surgery is an orthopedic procedure in which a surface of a diseased or a damaged joint is replaced with an orthopedic prosthesis designed to replicate the movement of a normal, healthy joint. In such orthopedic procedure, a patient's joint is typically replaced with prosthetic components by resecting a portion of the patient's bone and/or cartilage to create a platform, a recess, or a cavity for receiving a portion of the prosthetic components being implanted. Thereafter, the prosthetic components are affixed to the resected bone. However, in addition to replacing the joint surfaces of the prosthesis, some joint replacement surgeries may also require attachment of soft tissue to an underlying support structure, such as bone or the joint prosthesis itself.
(2) In one example, a revision joint replacement surgery typically involves the replacement of a primary prosthesis or a previously implanted revision prosthesis. Such surgery may additionally involve the correction of complications resulting from a joint replacement surgery. Such complications include suprapatellar, transpatellar, or infrapatellar extensor mechanism disruption which may occur either intra-operatively (e.g., as a result of improper patellar resection, damage to the blood supply due to injudicious lateral retinacular release, and the like) or in the immediate post-operative period (e.g., as a result of tissue necrosis arising from infection, component malalignment, and the like) of a total knee arthroplasty. Though non-surgical means may address the extensor mechanism disruption, based on the complexity and extent of the disruption, surgical intervention may be necessary.
(3) In another example, a significant amount of bone may be removed from an end of a patient's long bone. This may be due to a cancerous tumor, such as in the distal femur or proximal tibia, or perhaps due to severe trauma. In such a procedure, the connection points between soft tissue and bone are often removed along with the resected bone. For example, a tibial tubercle, which is the connection point for a patellar tendon, may be removed in one of these procedures. In this regard, the tendon is left with no natural connecting structures.
(4) In the above circumstances, along with others not mentioned, attachment of soft tissue (e.g.,

ligaments and tendons) necessary for functional operation of the corresponding joint may be difficult particularly where long term fixation of the tissue is required. In response to this difficulty, certain devices have been devised. For example, current prostheses that address soft tissue connection typically have openings in their solid structure which allows soft tissue to be sutured thereto by threading wiring, such as cerclage wire, or suture through the soft tissue and through the openings available on the prosthesis. However, such a configuration requires the suture or wires to maintain the connection of the soft tissue to the prosthesis over extended periods of time all the while forces are imposed on the soft tissue via normal joint movements. In this regard, if a single suture or wire fails, the connection between soft tissue and joint prosthesis may be compromised. Thus, further improvements are desirable.

BRIEF SUMMARY OF THE INVENTION

(5) Generally, disclosed herein are orthopedic assemblies and methods for attaching soft tissue to such orthopedic assemblies. In particular, such assemblies are each described as including a joint prosthesis and a filamentary attachment structure which connects to the joint prosthesis. The filamentary attachment structure allows soft tissue to be connected thereto and facilitates tissue ingrowth therein to facilitate a strong long-term connection.

(6) In one aspect of the present disclosure, an orthopedic assembly includes a tibial component having a baseplate, a stem, and a keel. The baseplate includes an upper articulating surface and lower surface disposed opposite the upper articulating surface. The stem and the keel extend from the lower surface, and the keel is positioned adjacent to the stem. The orthopedic assembly also includes a filamentary receiving component that includes a first portion and a second portion. The first portion includes an upper surface corresponding to the lower surface of the baseplate and an opening extending through the first portion which is configured to receive the stem and the keel therein. The second portion extends from an anterior end of the first portion.

(7) Additionally, when the stem is inserted into an intramedullary canal of a proximal tibia, the first portion of the filamentary receiving component may be positioned between the lower surface of the baseplate and the proximal tibia such that the first portion of the filamentary receiving component may be sandwiched therebetween. The upper articulating surface is configured to articulate with a femoral articulating surface, and the femoral articulating surface includes a lateral femoral condyle and a medial femoral condyle on a distal end of a femur. An anterior surface of the second portion of the filamentary receiving component may be configured to connect to soft tissue and includes a portion thereof that tapers away from the first portion. The second portion of the filamentary receiving component may include a filamentary material configured for securing soft tissue thereto. The filamentary material may be selected from the group consisting of: a synthetic polymer, a bioresorbable fiber, a ceramic/a glass, a biological material, a pharmacological agent, and combinations thereof.

(8) Continuing with this aspect, a thickness between the upper surface of the first portion and a lower surface of the first portion may be smaller than a distance from a perimeter of the first portion and the opening of the first portion. The opening may extend along a central axis. The central axis may extend perpendicular to the upper surface of the first portion of the filamentary receiving component. The keel may comprise at least two fins extending radially outward from the stem towards a posterior end of the baseplate. The opening may be defined by a central cylindrical portion and a lobe portion extending outwardly from the central cylindrical portion. The central cylindrical portion may be configured to receive the stem and the lobe portion may be configured to receive the keel. The lower surface of the baseplate and the upper surface of the filamentary receiving component may be planar.

(9) In another aspect of the present disclosure, a method for attaching soft tissue to a joint replacement assembly includes a joint replacement prosthesis and a receiving component includes inserting a stem of the joint replacement prosthesis into the receiving component. The receiving component includes a first portion and a second portion extending from the first portion, and the

joint replacement prosthesis includes an articular portion and a bone abutment portion disposed opposite the articular portion. The stem extends from the bone abutment portion. The method also includes inserting the stem of the joint replacement prosthesis into a bone so that the first portion of the receiving component is positioned between the bone abutment portion and the bone, and so that the second portion extends towards the articular portion of the joint replacement prosthesis. The method further includes securing soft tissue to the second portion of the receiving component.
(10) Additionally, the method may include, prior to inserting the stem of the joint replacement prosthesis into the receiving component, applying an adhesive substance to the bone abutment portion of the joint replacement prosthesis. The adhesive substance may be a bone cement. The second portion of the receiving component may include a filamentary material, and the filamentary material may be selected from the group consisting of: a synthetic polymer, a bioresorbable fiber, a ceramic/a glass, a biological material, a pharmacological agent, and combinations thereof. Also, the step of securing the soft tissue to the second portion of the receiving component may include suturing the soft tissue to the second portion. The second portion of the receiving component may include one or more suture holes. The second portion of the receiving component may be made from a knitted or woven filamentary material and the one or more suture holes may be defined by the knitted or woven filamentary material. Also, the step of securing the soft tissue to the second portion of the receiving component may include threading cerclage wire through the one or more suture holes of the second portion and through the soft tissue. The method may further include threading the cerclage wire through openings in the joint replacement prosthesis.
(11) Continuing with this aspect, the method may further include securing the soft tissue directly to the joint replacement assembly via a threaded connection or a wired connection. The bone may be a tibia, and the soft tissue may be a patellar tendon.
(12) In a further aspect of the present disclosure, an orthopedic assembly includes a tibial component that includes a baseplate, a stem, and a keel. The baseplate includes an upper articulating surface and a planar lower surface disposed opposite the upper articulating surface. The stem and the keel extending from the planar lower surface, and the keel includes a first fin positioned adjacent to the stem. The assembly also includes a filamentary receiving component that includes a first portion and a second portion. The first portion includes an upper surface, a lower surface, and an opening extending through the upper and lower surfaces. The opening includes a main region and a first offset region in communication with the main region. The main region is configured to receive the stem, and the first offset region is configured to receive the first fin.
(13) Additionally, the second portion may include an attachment region located on an anterior surface of the second portion. The attachment region may include a filamentary material configured to secure a soft tissue thereto. The attachment region may further include one or more porous surfaces configured to facilitate ingrowth of a bone and/or the soft tissue. The filamentary receiving component may also include a second offset region in communication with the main region. The keel may include a second fin positioned adjacent to the stem, and the second offset region may be configured to receive the second fin. The first offset region may be positioned on a first side of a central axis extending through the main region of the opening, and the second offset region may be positioned on a second side of the central axis. The main region may be cylindrical, and the first and the second offset regions may be elongate and extend away from the main region in both an anteroposterior direction and a mediolateral direction.
(14) In an additional aspect of the present disclosure, an orthopedic assembly includes a tibial prosthesis that includes a body that defines an anterior side and a posterior side. The body further defines a recess in the anterior side of the joint prosthesis and a plurality of openings extending through the body from the anterior side to the posterior side thereof. At least a first and second opening of the openings are positioned at respective lateral and medial sides of a longitudinal axis of the tibial prosthesis. The assembly also includes a modular insert positioned within the recess of the body such that at least a portion of the modular insert is positioned between the openings of the

body. The modular insert is formed separately from the tibial prosthesis and has a porous outer surface to promote tissue ingrowth.
(15) Additionally, the modular insert may include a ring portion positioned about a stem of the body and a connection portion extending from the ring portion and positioned within the recess of the body. The connection portion may include at least one protrusion extending therefrom and into a complementary receiving structure of the body. The complementary receiving structure may be located within the recess of the body. The at least one protrusion may be tapered and the complementary receiving structure may be tapered so as to form a taper lock when the protrusion is received in the complementary receiving structure.
(16) Continuing with this aspect, the assembly may also include a filamentary fixation device extending through the first and second openings. The filamentary fixation device may have a first free end and a second free end. The first and second free ends may extend through the respective first and second openings such that the first and second free ends are positioned at the anterior side of the body. The filamentary fixation device may include a filamentary material configured to secure soft tissue thereto. The filamentary material may be selected from the group consisting of a synthetic polymer, a bioresorbable fiber, a ceramic/a glass, a biological material, a pharmacological agent, and combinations thereof. The filamentary fixation device may include a mesh structure for securing soft tissue thereto. The mesh structure may include a plurality of layers. The plurality of layers may be connected by a continuous or interrupted seams. A third opening of the openings in the body may extend in a lateral-medial direction for receipt of a suture therethrough to connect the filamentary fixation device to the body.
(17) Furthermore, the assembly may include cannulated screws. The cannulated screws may connect the modular insert to the body via the first and second holes. The cannulated screws are cannulated such that they each include an opening extending entirely therethrough such that when they are received in the first and second openings of the body, such openings are not entirely occluded by the cannulated screws.
(18) In a further aspect of the present disclosure, an orthopedic assembly includes a joint prosthesis that includes a body. The body has an upper portion and a stem. The upper portion has a recessed portion that has at least one opening extending through the body. The assembly also includes a plate separately formed from the joint prosthesis and has a shape complementary to and received by the recessed portion of the body. The plate has at least one opening extending therethrough and a porous outer surface. The at least one opening of the plate is aligned with the opening of the body when the plate is received by the recessed portion. The assembly further includes a threaded member. The threaded member is cannulated such that an opening extends through the length of the threaded member. The threaded member connects the plate to the joint prosthesis via the openings of the body and plate such that the cannulation of the threaded member prohibits such openings from being completely occluded when the threaded member is received therein.
(19) Additionally, the opening of the threaded member may be configured to receive a suture or a wire. The plate may include an attachment region at an anterior side thereof. The attachment region may include the porous outer surface configured to facilitate the ingrowth of a soft tissue. The porous outer surface may be made from a metallic material. The plate further may further include an indented region at an anterior side thereof for receipt of a segment of bone. The at least one opening of the plate and joint prosthesis may include a first and second opening and one of the threaded member may be disposed in each of the first and second openings of the plate and joint prosthesis to fix the plate to the joint prosthesis.
(20) Continuing with this aspect, the assembly may include a suture or a wire extending through the first and second openings of the plate, joint prosthesis, and respective threaded members thereof. The joint prosthesis may be a tibial prosthesis having a proximal end configured to articulate with a femoral prosthesis. The tibial prosthesis may include a metaphyseal portion and a diaphyseal portion for replacing a respective metaphysis and diaphysis of bone.

(21) In an even further aspect of the present disclosure, an orthopedic system includes a joint prosthesis that has a first portion configured to connect to a resected bone and a second portion configured to interface with a second joint prosthesis. The system also includes a filamentary fixation device for reconstructing soft tissue that has a plurality of layers of a mesh material. Each of the layers is connected to an adjacent layer by at least one seam formed by heat.
(22) Additionally, the plurality of seams may be continuous along the length of the filamentary fixation device. The plurality of seams, alternatively, may be discontinuous along the length of the filamentary fixation device such that the seam extends the full length of the filamentary fixation device but has free segments disposed at regular intervals along the axis of the seam.
(23) In yet further aspect of the present disclosure, an orthopedic assembly includes a tibial prosthesis that includes a body that defines an anterior side and a posterior side. The body further defines first and second slots extending through the posterior and anterior sides of the body. The first and second slots each define a posterior and anterior aperture. The anterior aperture includes a triangular shape. The orthopedic assembly also includes a filamentary fixation device extending through the first and second slots.
(24) Additionally, the posterior aperture may have opposing parallel sides. The posterior aperture may be pill-shaped. Each of the first and second slots may be defined by a first and second sidewall. The first sidewall may tilt away from a central longitudinal axis of the body. The first sidewall may form a hypotenuse of the triangular shaped anterior aperture. The first sidewall may be parallel to the central longitudinal axis at the posterior aperture and may gradually tilt away from the central longitudinal axis along a traversal of the first sidewall from the posterior aperture to the anterior aperture. The first sidewall may decrease in a superior-inferior height along its traversal from the posterior aperture to the anterior aperture. The second sidewall may maintain a constant superior-inferior height along its traversal from the posterior aperture to the anterior aperture. The first sidewall may curve about the central longitudinal axis of the body.

Description

BRIEF DESCRIPTION OF THE FIGURES
(1) A more complete appreciation of the subject matter of the present invention and the various advantages thereof may be realized by reference to the following detailed description, in which reference is made to the following accompanying drawings:
(2) FIG. **1** is a bottom perspective view of a joint replacement assembly according to an embodiment of the present disclosure.
(3) FIG. **2** is a top perspective view of the joint replacement assembly of FIG. **1**.
(4) FIG. **3** and FIG. **4** are exploded views of the joint replacement assembly of FIG. **1**.
(5) FIG. **5** is a side perspective view of a joint replacement assembly according to another embodiment of the present disclosure.
(6) FIG. **6** and FIG. **7** are exploded views of the joint replacement assembly of FIG. **5**.
(7) FIG. **8** is a side perspective view of a joint replacement assembly according to a yet further embodiment of the present disclosure including a joint replacement prosthesis and a tissue attachment structure.
(8) FIG. **9** is a front perspective view of the joint replacement prosthesis of FIG. **8**.
(9) FIG. **10**A and FIG. **10**B are perspective views of the tissue attachment structure of FIG. **8**.
(10) FIG. **11** is a side perspective view of a joint replacement assembly according to yet another embodiment of the present disclosure.
(11) FIG. **12** is a front perspective view of a tissue attachment structure according to an even further embodiment of the present disclosure.
(12) FIG. **13** is a schematic representation of a weave of filamentary material according to an

embodiment of the present disclosure.
(13) FIG. **14**A is an exploded view of a joint replacement assembly according to an even further embodiment of the present disclosure.
(14) FIG. **14**B is a rear perspective view of the joint replacement assembly of FIG. **14**A in a first configuration.
(15) FIG. **14**C is a front perspective view of the joint replacement assembly of FIG. **14**A in the first configuration.
(16) FIG. **14**D is a top perspective view of the joint replacement assembly of FIG. **14**A in a second configuration.
(17) FIG. **14**E is a front perspective view of the joint replacement assembly of FIG. **14**A in a second configuration.
(18) FIG. **15**A is an exploded view of a joint replacement assembly according to another embodiment of the present disclosure.
(19) FIG. **15**B is a rear perspective view of the joint replacement assembly of FIG. **15**A.
(20) FIG. **15**C is a front perspective view of the joint replacement assembly of FIG. **15**A.
(21) FIG. **16** is a front perspective view of a joint replacement assembly according to yet another embodiment of the present disclosure.
(22) FIG. **17** is a front perspective view of a joint replacement assembly according to a still further embodiment of the present disclosure.
(23) FIG. **18**A is a front elevational view of a joint replacement prosthesis according to another embodiment of the present disclosure.
(24) FIG. **18**B is a top view of the joint replacement prosthesis of FIG. **18**A.
(25) FIG. **18**C is a partial rear elevational view of the joint replacement prosthesis of FIG. **18**A.
(26) FIG. **18**D is a rear perspective view of a joint replacement assembly including the joint replacement prosthesis of FIG. **18**A and a filamentary fixation device of FIG. **14**A.
(27) FIG. **18**E is a partial front perspective view of the joint replacement assembly of FIG. **18**D.
(28) FIG. **19**A is a schematic perspective view of a stack of filamentary material.
(29) FIG. **19**B is a top view of the stack of filamentary material of FIG. **19**A.
(30) FIG. **19**C is a top view of a filamentary fixation device according to a further embodiment of the present disclosure including the filamentary material of FIG. **19**A.
(31) FIGS. **19**D and **19**E are schematic cross-sectional views taken along line A-A of FIG. **19**C.
(32) FIG. **19**F is a top view of another filamentary fixation device including the stack of filamentary material of FIG. **19**A.
(33) FIG. **20**A is a front perspective view of a joint replacement prosthesis including a connection plate and according to an even further embodiment of the present disclosure.
(34) FIG. **20**B is a rear elevational view of the joint replacement prosthesis of FIG. **20**A.
(35) FIG. **20**C is a rear perspective view of the connection plate of FIG. **20**A.
(36) FIG. **20**D is rear perspective view of the connection plate of FIG. **20**A including a fastener assembled therewith.
(37) FIG. **20**E is a cutaway perspective view of the joint replacement prosthesis of FIG. **20**A.
(38) FIG. **20**F is a side cross-sectional view taken along line F-F of FIG. **20**A.
DETAILED DESCRIPTION
(39) When referring to specific directions in the following discussion of certain implantable joint replacement devices, it should be understood that such directions are described with regard to the orientation and position of the implantable joint replacement devices during exemplary application to the human body. Thus, as used herein, the term “proximal” means situated nearer to the center of the body or the point of attachment and the term “distal” means more situated away from the center of the body or from the point of attachment. The term “anterior” means towards the front part of the body or the face and the term “posterior” means towards the back of the body. The term “medial” means toward the midline of the body and the term “lateral” means away from the

midline of the body. Further, as used herein, the terms “about,” “generally,” and “substantially” are intended to mean deviations from absolute are included within the scope of the term so modified.
(40) FIGS. **1-4** depict a joint replacement assembly **100** comprising a joint replacement prosthesis **102** and a filamentary fixation device **120**. The joint replacement prosthesis **102** is a tibial component of a total knee arthroplasty system and is made from biologically suitable material for implantation, such as titanium, stainless steel, cobalt chromium, niobium, and the like. In addition, joint replacement prosthesis **102** may have porous outer surfaces to facilitate bone ingrowth therein.
(41) Joint replacement prosthesis **102** comprises a tibial baseplate **106**, a stem boss **104**, and a keel **112**. Tibial baseplate **106** comprises an insert mating portion **108** and a planar lower surface **110** disposed opposite insert mating portion **108**. Planar lower surface **110** is configured to engage a proximal sub-condylar area of a tibia, which is formed by resecting a proximal tibia transverse to an axis thereof. Moreover, planar lower surface **110** may further comprise engagement features (not shown), such as a porous or corrugated surface, or a rim depending downwardly therefrom about a perimeter thereof for engaging the proximal sub-condylar area of the tibia. However, surface **110** is generally understood to be planar.
(42) Stem boss **104** and keel **112** extend from planar lower surface **110**. Keel **112**, which appears as a wing or blade, is configured to prevent rotation of tibial baseplate **106** and is positioned adjacent stem boss **104**. In this regard, keel **112** has a length that extends in a lateral-medial direction, but may also include a component of its travel in an anteroposterior direction. In the particular configuration depicted, prosthesis **102** includes two keels **112***a-b*, which are integral with baseplate **106** and stem boss **104**. However, in some embodiments, only one keel **112** may be provided. Also, keels **112***a-b* may be modular such that they are not rigidly fixed to either lower surface **110** or stem **104**. In still further embodiments, keels **112***a-b* may be connected to lower surface **110**, but not stem boss **104**. Regardless, in each of these embodiments, keels **112***a-b* extends from lower surface **110** and are positioned adjacent stem **104**. Stem boss **104** is cylindrical or frustoconical in shape to substantially match a cavity of a bone, such as an intramedullary canal of the tibia. In addition, stem boss **104** is configured to connect to a stem extension (not shown), such as via an opening **114** in stem boss **104**.
(43) As best shown in FIG. **2**, insert mating portion **108** of tibial baseplate **106** is defined by a shoulder portion **107** (e.g., a rim) extending about a periphery of baseplate **106**. Such shoulder portion **107** forms a dish or tray configured to receive a tibial bearing component (not shown) where such bearing component includes an articular surface that is configured to articulate with a femoral component, as is understood in the art.
(44) Filamentary receiving component, filamentary fixation device, or filamentary sleeve **120**, as best shown in FIGS. **3** and **4**, includes a first portion **128** and a second portion **122** extending from an anterior end of first portion **128**. First portion **128** has a planar upper surface **125** that corresponds to planar lower surface **110** of tibial baseplate **106**. First portion **128** further comprises an opening **129** that extends through first portion **128**. A central axis (dotted line in FIG. **3**) extends perpendicular to planar upper surface **125** of first portion **128**. Opening **129** extends along the central axis and is symmetrical about an anterior-posterior plane in which the central axis lies. A thickness between planar upper surface **125** and a lower surface **124** of first portion **128** is smaller than a distance from a perimeter of first portion **128** and opening **129** of first portion **128**.
(45) Opening **129** comprises a main region **121** having a cylindrical shape and lobe portions or offset regions **123***a-b* extending outwardly from main region **121**. Main region **121** is sized and shaped to receive stem boss **104**, and lobe portions **123***a-b* are sized and shaped to receive keels **112**. In this regard, lobe portions **123***a-b* each extend radially outward from main region **121** both laterally-medially and anteriorly-posteriorly at an angle towards a posterior side of first portion **128** just as keels **112***a-b* extend from stem boss **104**. However, in some embodiments, first lobe **123***a* and second lobe **123***b* may extend away from main region **121** in only a mediolateral direction

depending on the configuration of keels **112***a-b*. A shape of first lobe **123***a* and second lobe **123***b* is not limited to the depicted and described shape as additional shapes are contemplated. However, the shape of lobes **123***a-b* generally correspond to a geometry of keels **112***a-b* so as to conform thereto.

(46) As mentioned above, second portion **122** extends from an anterior side of first portion **128**. As shown, second portion **122** also extends in a direction transverse to planar upper surface **125** and away from planar upper surface **125**. Also, as shown, second portion **122** has a smaller lateral-medial width than that of first portion **128**. In this regard, second portion **122** includes a transition region **127** (see FIG. **4**) in which second portion **122** tapers to its more narrow width from first portion **128**. Second portion **122** is also generally rectangular in a lateral medial plane and has a larger dimension in a lateral-medial direction than an anteroposterior direction.

(47) Filamentary receiving component **120** is made from a filamentary material that may be a knitted or woven material, a non-woven material, or a combination thereof. Such filamentary material may comprise one or more of: a synthetic polymer, a bioresorbable fiber, a ceramic/a glass, a biological material, and a pharmacological agent, among others. The synthetic polymer comprises one or more materials, such as: an ultra-high molecular weight polyethylene (UHMWPE), a polyether-ether-ketone (PEEK), a carbon reinforced PEEK, a polyether-ketone (PEK), a texturized polyethylene terephthalate (PET), an open-weave PET, and a polytetrafluorethylene (PTFE), among others. The bioresorbable fiber comprises one or more materials, such as: a polylacatic acid (PLA), a polyglycolide (PGA), and a poly-L-lactic acid (PLLA), among others. The ceramic/the glass comprises one or more of: an alumina, a zirconia, and a pyrolytic carbon, among others. The biological material comprises one or more materials, such as: a collagen, a silk, and a chitosan, among others. According to some embodiments, the filamentary material of filamentary receiving component **120** comprises a knitted or woven mesh material, such as a monofilament mesh material. It should be appreciated that the listed materials are non-exhaustive and other materials are contemplated herein. However, it should be understood that it is preferable that filamentary receiving component **120** be made from a material that encourages soft tissue growth therein. Thus, a knitted or woven material that has a weave that encourages tissue growth into its porous structure is preferable. An exemplary weave is shown in FIG. **13**.

(48) As assembled, filamentary receiving component **120** receives tibial component **102** to form joint replacement prosthesis **100**. In this regard, main portion **121** of opening **129** receives stem **104** of tibial baseplate **102**, first lobe **123***a* receives a first keel **112***a*, and second lobe **123***b* receives a second keel **112***b*. Moreover, planar lower surface of baseplate **106** sits generally flush against upper planar surface **125** of first portion **128**. The shape of first portion **128** corresponds to the shape of baseplate **106** so that first portion **128** does not extend beyond its perimeter with the exception of near the anterior side thereof where first portion **128** meets second portion **102**. In addition, first portion **122** extends upwardly toward mating portion **108** of baseplate **106**, as best shown in FIG. **2**. First portion **122** extends along the anterior surface of baseplate portion and beyond baseplate **106**.

(49) In a method for attaching the soft tissue to the joint replacement prosthesis **102**, a previously implanted joint prosthesis may be removed from a proximal tibia. After the proximal tibia is prepared, such as by resecting the proximal tibia, joint replacement assembly **100** is assembled and connected to the tibia. In this regard, filamentary receiving component **120** is engaged to joint prosthesis **102** by inserting stem boss **104** through main portion **121** of opening **129** and keels **122***a-b* through respective lobe portions **123***a-b* of opening **129** so that planar lower surface **110** of baseplate **106** is brought into communication with planar upper surface **125** of filamentary receiving component **120**. Thereafter, bone cement, such as polymethyl methacrylate (PMMA), is placed on a proximal surface of the tibia and/or on lower surface **124** of first portion **128** of filamentary receiving component **128**. In some embodiments, bone cement may even be placed

between first portion of filamentary component and baseplate. As shown in FIG. **1**, a portion of lower surface **110** of baseplate **106** is not covered by first portion **128**. This portion of baseplate **106** also receives bone cement. Once the bone cement is applied, stem boss **104**, along with any stem extension attached thereto, is inserted into the intramedullary canal of the tibia and keels **112***a-b* are driven into the proximal end of the bone until baseplate **106** and filamentary component **120** engage the proximal end of the tibia. Once baseplate **106** is fully seated, first portion **122** of filamentary component **120** is trapped or sandwiched between baseplate **106** and the proximal tibia. Moreover, second portion **122** extends from between the proximal tibia and baseplate **106** and extends superiorly away from the tibia. In this configuration, second portion **122** is well secured by first portion's arrangement between bone and baseplate **106** and also around stem boss **104** and keels **112***a-b*.
(50) Once assembly **100** is mounted to the tibia, soft tissue is secured to joint replacement prosthesis **102** via filamentary device **120**. In this regard, an intact patellar tendon can be attached to filamentary device **120**, such as in a revision procedure, to reinforce the tendon from subsequent damage as patellar tendon tears regularly occur postoperatively. Alternatively, a patellar tendon may have been detached from the tibia for any number of reasons. In order to re-secure the intact or detached patellar tendon, the patellar tendon is sewn to a posterior side of second portion **122**. Moreover, a muscle, such as the medial gastrocnemius may be sewn to an anterior side of second portion **122** of filamentary component **120** via suture or wire. The arrangement of the patellar tendon and muscle is partially illustrated in FIG. **8**. This configuration allows the soft tissue to grow into filamentary component **120** thereby providing a strong long term connection. Thus, filamentary component **120** provides a soft tissue ingrowth structure to prosthesis **102**.
(51) FIGS. **5-7** depict a joint replacement assembly **200** according to another embodiment of the present disclosure. Joint replacement assembly **200** includes joint replacement prosthesis **201** and filamentary receiving component or filamentary fixation device **220**.
(52) Joint replacement prosthesis **201** is a tibial prosthesis. However, unlike prosthesis **102** which is a revision tibial prosthesis, prosthesis **201** is configured for use in limb salvage procedures, such as for oncology applications. In this regard, prosthesis **201** comprises a baseplate component **202** that includes a tibial baseplate **212** and a stem **204** extending from tibial baseplate **212**. Prosthesis **201** also includes a separately formed body member or metaphyseal member **240** which is connectable to baseplate component **202**.
(53) Tibial baseplate **212** comprises an insert mating portion **206** and a planar lower surface **214** disposed opposite insert mating portion **206**. Insert mating portion **206** is configured to articulate with a femoral articulating surface on a distal end of a femur. In this regard, insert mating portion **206** may be configured to receive a bearing insert (not shown) similar to baseplate **106**, described above. Planar lower surface **214** includes projections or posts **216** extending from a lateral and medial side thereof. Such posts are configured to be received in apertures **246** of body member **240**, as discussed below. Stem **204** extends from planar lower surface **214** of tibial baseplate **212** and includes a connection feature **205** at a distal end thereof for connecting to a resected tibial shaft. In this regard, stem **204** acts as a diaphyseal portion of prosthesis **201** for replacing a portion of a diaphysis of a tibia.
(54) Body member **240** is a metaphyseal portion for replacing a metaphysis of a tibia and includes a proximal end, a distal end, and an opening **247** that extends therethrough for receipt of stem **204**. The proximal end of body **240** includes a planar surface **248** and apertures **246** extending therein for receipt of posts **216**, as best shown in FIG. **6**. Such apertures **246** and posts **214** may be correspondingly tapered, such as to form Morse taper locks. An outer surface of body member **240** is tapered so that body **240** expands outwardly in a distal to proximal direction. Additionally, one or more attachment holes **242** and an attachment region **244** are located on the anterior side of body member **240** and are configured for securing a soft tissue to joint replacement assembly **200**. In this regard, attachment holes **242** extend in a generally lateral-medial direction through body member

240 so that such attachment holes **242** can receive suture, wire, and the like. Attachment region **244** may also include a patch of filamentary material embedded in the solid structure of body member **244**, which can allow soft tissue to be sutured and/or can facilitate tissue growth therein.

(55) Filamentary receiving component **220** is similar to filamentary receiving component **120**. In this regard, filamentary receiving component **220** includes a first portion **229** and a second portion **222** extending from first portion **229**. First portion **229** includes upper and lower planar surfaces **225**, **224** and an opening extending therethrough. The opening includes a main region **221** and adjacent lobe regions **223** similar to those of component **120**. However, adjacent lobe regions **223** are configured to receive projections rather than keels **112**. Thus, while lobes **223** are shown as communicating with main region **221**, such as with component **120**, it is contemplated that lobes **223** may not be in communication with main region **221** and, instead, may be positioned remote from main region **221** in respective locations to receive projections **216**.

(56) As assembled, stem **204** extends through main region **221** of the opening extending through first portion **229** and through the opening **247** in body member **240**, as best illustrated in FIGS. **6** and **7**. In addition, projections **216** extend through corresponding lobes **223** and into corresponding apertures **246** in body member **240**, which secures body member **240** to baseplate **212**. In this configuration, first portion **229** of filamentary receiving component **220** is trapped between planar surface **248** of body member **240** and baseplate **212**. This is similar to the finally implanted assembly **100** in which first portion **229** is trapped or sandwiched between baseplate **106** and a proximal tibia. However, in this configuration, first portion **229** is trapped or sandwiched between baseplate **212** and body member **240**, rather than between bone and a prosthetic component. In addition, second portion **222** extends from between baseplate **212** and body member **240** and superiorly beyond baseplate **212**.

(57) In a method for attaching the soft tissue to joint replacement prosthesis **201**, a patellar tendon is detached from the tibial tubercle and a proximal section of the tibia is resected at a location along the tibial shaft so that the removed bone includes the tibial tubercle. Thereafter, filamentary receiving component **220** is engaged to joint prosthesis **201** by inserting stem **204** through opening main region **221** of the opening in first portion **229** and projections **216** extend through lobes **223** into corresponding apertures **246** so that first portion **229** is trapped between surface **248** of body **240** and surface **214** of baseplate **212**. Implant **201** is then connected to the bone via connection portion **205**.

(58) Once assembly **200** is mounted to the tibia, soft tissue, such as a patellar tendon, is secured to joint replacement assembly by suturing the soft tissue to second portion **222** of filamentary receiving component **220** as described above with respect to assembly **100**. The soft tissue may also be secured to joint replacement prosthesis **201** directly by threading wire, such as cerclage wiring, or suture through the soft tissue and through one or more suture holes **242** on joint replacement assembly **200**.

(59) FIGS. **8-10**B depict a joint replacement assembly **300** according to an even further embodiment of the disclosure. Joint replacement assembly **300**, as best shown in FIG. **9**, comprises a joint prosthesis **302** and a filamentary receiving component or filamentary fixation device **340**. Joint prosthesis **302** is a tibial component of a hinge knee system that may be used in limb salvage procedures, such as for oncology applications. In this regard, prosthesis includes a body **330**, a bearing portion **320**, and a hinge portion **310** that bears on bearing portion **320**. Body **330** has a diaphyseal portion **336** and a metaphyseal portion **334** for replacement of the same of a tibia. At a distal end of diaphyseal portion **336** is a connection feature **338** that allows prosthesis **302** to be connected to a resected portion of a tibial shaft.

(60) Metaphyseal portion **334** includes one or more suture holes or openings **332** located on a medial side and/or a lateral side of an anterior face thereof and extending entirely therethrough. According to some embodiments, a region between suture holes **332** may comprise a material integrated into the structure of body **330** so that body **330** has a patch of filamentary material

embedded therein to assist in securing the soft tissue to joint replacement assembly **300** and/or one or more porous surfaces to support tissue ingrowth. Metaphyseal portion **334** is connected to a proximal end of diaphyseal portion **336** so that body **330** tapers outwardly in a distal to proximal direction.

(61) Filamentary receiving component **340**, as shown in FIGS. **10**A and **10**B, is similar to filamentary receiving component **120** and **220** in that it is made from filamentary material and includes a first portion **344** and second portion **342**. Moreover, first portion **344** has an opening **346** extending therethrough and second portion **342** extends from an anterior side of first portion **344**. However, unlike filamentary receiving components **120** and **220**, filamentary component **340** has an annular opening **346** that flares or tapers outwardly in an inferior to superior direction so as to form an annulus with a sidewall that is thinner than the proximal-distal length of second portion. This allows filamentary component **340** to correspondingly engage a prosthesis, such as prosthesis **302**, which also tapers along its length. Thus, in the assembly, as shown in FIG. **8**, diaphyseal portion **336** extends through opening **346** in first portion **344** of filamentary component **340**. The inner surface of first portion **346** bears on body **330** along the tapered outer surface thereof such that the taper of body **330** prohibits second portion **344** from moving proximally beyond a predesignated location on body **330**.

(62) In a method for attaching the soft tissue to joint replacement prosthesis **302**, a patellar tendon **902** is detached from the tibial tubercle and a proximal section of the tibia is resected at a location along the tibial shaft so that the removed bone includes the tibial tubercle. In addition, prosthesis **302** is assembled by connecting body **330**, bearing **320**, and hinge component **310**. Thereafter, filamentary receiving component **340** is engaged to joint prosthesis **302** by inserting diaphyseal portion **336** through opening **346** so that the inner surface of first portion **344** is brought into communication with a correspondingly tapered outer surface of prosthesis **302**. Implant **302** is then connected to the bone via connection portion **338**.

(63) Once assembly **300** is mounted to the tibia, soft tissue is secured to joint replacement prosthesis **302** via filamentary device **340**. In order to re-secure the detached patellar tendon **902**, the patellar tendon **902** is sewn to a posterior side of second portion **342**, as depicted in FIG. **8**. Moreover, a muscle **904**, such as the medial gastrocnemius may be sewn to an anterior side of second portion **342** of filamentary component **340** via suture or wire. In addition, a suture **906** may be threaded through openings **332** and passed through tendon **902**, first portion **342** of filamentary component **340**, and muscle **904**. Thus, this configuration allows for a traditional connection to implant **302** via openings **332** and also allows for the soft tissue **902**, **904** to grow into filamentary component **340** thereby providing a strong long term connection. Thus, filamentary component **340** provides a soft tissue ingrowth structure to prosthesis **302**.

(64) FIG. **11** depicts a joint replacement assembly **400** according to yet another embodiment of the present disclosure. Joint replacement assembly **400** generally includes a joint replacement prosthesis **402** and a filamentary receiving component or filamentary fixation device **440**. Joint replacement prosthesis **402** is a humeral joint prosthesis. Thus, as should be understood, the concepts described above with regard to the tibial prostheses embodiments may also be applied to other long bones and joints, such as the humerus and shoulder joint in this embodiment. It should also be understood that the same concepts could also apply to a joint prosthesis for proximal femur and hip joint, for example. Such a hip joint prosthesis would be similarly configured to prosthesis **402** in that it would include an intramedullary stem and a ball joint portion at an end thereof.

(65) Prosthesis **402** comprises a proximal end remote from a distal end. The proximal end includes a head portion **412** that includes a bearing surface that articulates with a glenoid component (not shown) when used in a total should replacement system. Head **412** is connected to a body element **418** which connects to a stem portion **404**. Stem portion **404** comprises a tip portion **403** that is adapted to be inserted into an intramedullary canal of a proximal humerus such that stem portion **404** is located within the intramedullary canal thereof. Body element **418** is preferably conically

tapered so as to correspond to a taper of an opening of filamentary component **440**, as described below. Filamentary receiving component **440** is the same as filamentary receiving component **340** and may be used similarly. In this regard, stem **404** is inserted through an opening in a first portion **444** of filamentary component **440** so that filamentary component **440** bears on body **418** and is retained by corresponding tapers. Additionally, a first portion **442** extends superiorly beyond head element **412**. Moreover, a tendon, such as one or all of the tendons of the rotator cuff, may be connected to first portion of filamentary device **440** via suture or wire. In this regard, tissue may grow into first portion **442** to provide long term fixation thereof to prosthesis **402**.

(66) FIG. **12** depicts a front perspective view of an alternative filamentary fixation device **500** configured for use with a joint replacement prosthesis or a joint replacement assembly, according to at least some embodiments disclosed herein. Filamentary device **500** comprises a hoop structure **503** and an opening **501** located within hoop structure **503**. Hoop structure **503** is defined by arm members **504***a* and **504***b*. Such arm members **504***a-b*, as shown, are connected together to form hoop structure **503**. However, arm members **504***a-b* are preferably provided disconnected so that they may be tied together or otherwise connected during a procedure, such as about a stem boss **104** and keels **112** or about stem **336** or **403**, for example. Moreover, arm members **504***a-b* may be each comprised of single or double strands of filament such that they have an appearance of shoelaces. This differs from first portion of filamentary device **120** in that when device **500** is connected to prosthesis **102**, its footprint relative to baseplate **106** is much smaller than that of filamentary device **120**. This allows hoop structure to be trapped between bone and a prosthesis or between components of a prosthesis, as described above, while having a low profile so as to not interfere with direct contact between the bone and prosthesis or prosthesis and prosthesis.

(67) FIGS. **14**A-**14**E depict a joint replacement assembly **600** according to yet another embodiment of the disclosure. Joint replacement assembly **600** is similar to joint replacement assembly **300** in that it includes a joint prosthesis **602** similar to joint prosthesis **302**. In this regard, joint prosthesis **602** includes a diaphyseal portion **605** with a connection feature **604** configured to connect to a resected portion of a tibial shaft and a metaphyseal portion **634** configured to interface/articulate with another joint prosthesis, such as a femoral prosthesis (not shown). In the particular embodiment depicted, metaphyseal portion is particularly configured to form a hinged connection with a femoral component. In this regard, metaphyseal portion includes a tray region **642** for a tibial insert (not shown) and an elongate opening **641** extending therein for a stem of an axle assembly (not shown). Exemplary components that can be used in conjunction with assembly **600**, and other tibial assemblies described herein, can be found in U.S. Pub. No. 2017/0035572, the disclosure of which is hereby incorporated by reference herein in its entirety.

(68) However, assembly **600** differs in that it also includes a filamentary fixation device **640** and a connectable sleeve **662**. Moreover, joint prosthesis **602** is configured to receive filamentary fixation device **640** and connectable sleeve **662**. In this regard, metaphyseal portion **634** includes openings or elongate slots **658** which extend through metaphyseal portion **634** in an anteroposterior direction and are positioned at opposite lateral and medial sides of a longitudinal axis of prosthesis **602**. Such elongate slots **658** have their longitudinal length extending in a generally superior-inferior direction. This elongate configuration corresponds to a flat geometry of filamentary fixation device **640**, as described below.

(69) Metaphyseal portion **634** also includes one or more openings **632** located at an anterior side **634***b* of prosthesis **602** and extends entirely through prosthesis **602** in a lateral-medial direction. Another opening **660** similarly extends through prosthesis **602** in a lateral-medial direction. In addition, opening **660** intersects elongate slots **658** so as to be in communication therewith, as best shown in FIG. **14**A. The intersection of opening **660** with elongate slots **658** allows for a suture or the like to be threaded through filamentary fixation device **640** while disposed in slots **658** and through opening **660** in order to help retain device **640** in a connected state with prosthesis **602**.

(70) As mentioned above, prosthesis **602** is also configured to receive sleeve **662**. In this regard,

diaphyseal portion **605** is conically tapered so as to accommodate sleeve **662** in a taper lock configuration. In addition, metaphyseal portion **634** includes an anterior recess **669** that extends into the anterior face of metaphyseal portion **634**. Anterior recess **669** intersects or communicates with openings **632** and **660** to form longitudinal grooves **667** and **668** that are located within the perimeter of recess **669**.

(71) Sleeve **662**, as best shown in FIG. **14**A, includes a ring or cylindrical portion **661** and a tab portion or ingrowth portion **663** extending superiorly therefrom. Ring portion **661** has an opening **666** extending therethrough and is correspondingly tapered relative to portion **605**, as indicated above. Also, as mentioned above, tab portion **663** includes projections **664** which extend posteriorly for communication with grooves **667** and/or **668**. In this regard, projections **664** may have a concave surface (not shown) that extends laterally-medially to form a groove that corresponds to that of grooves **667** and **668**. Thus, when tab portion **663** is positioned within anterior recess **669**, projections **664** mate with grooves **667**/**668** so that the concave surfaces of projections **664** and grooves **667**/**668** together form cylindrical channels extending through prosthesis **602**. Preferably when tab portion **663** is received within recess **669**, an outer surface thereof is flush with an outer surface of prosthesis **602**. Preferably, tab portion **663** of sleeve **662** is made of a porous material, such as titanium foam, to support tissue ingrowth. For example, an exterior surface opposite projections **664** may be porous while projections **664** may be made from a solid metal material. Such configuration may be made via an additive manufacturing process so as to form a unitary structure. In addition, ring portion **661** may also be made of the same porous material or conversely a solid structure. Although the embodiment depicted includes a taper-lock mechanism to connect sleeve **662** to prosthesis **602**, other locking mechanisms are contemplated. For example, one or more threaded fasteners may connect ring portion **661** to diaphyseal portion **605**.

(72) Filamentary fixation device **640** is similar to filamentary fixation device **120** in that it is made from a filamentary material that may be a knitted or woven material, a non-woven material, or a combination thereof. However, filamentary fixation device **640** is an elongate flat strip of filamentary material that, when initially connected to prosthesis **602**, is folded over so that it resembles “U” shape with an inner surface **640***a* and an outer surface **640***b*.

(73) Thus, as assembled, terminal ends **643** of filamentary fixation device **640** are threaded through elongate slots **658** in a posterior to an anterior direction (see FIGS. **14**D and **14**E) so that inner surface **640***b* of the U-shaped construct engages a posterior side **634***a* of metaphyseal portion **634**. The flat profile of filamentary fixation device **640** helps it conform to prosthesis **602** to provide a low profile. Thus, filamentary fixation device **640** has free ends **643** that can wrap around the anterior outer surface **634***a* of metaphyseal portion **634** with the ends **643** of filamentary fixation device **640** going through respective openings **658**. A suture (not shown) may be passed through suture opening **660** and fixation device **640** at both the lateral and medial sides of prosthesis **602** to further secure filamentary fixation device **640** and to prosthesis **602** and to prevent device **640** from backing out of openings.

(74) Also, as assembled, ring portion **661** of sleeve **662** receives diaphyseal portion **605** so that tab portion **663** extends superiorly and so that an outer surface of portion **605** interferes with an inner surface of ring portion **661** to form a taper lock. Projections **664** of tab portion **663** are positioned in communication with grooves **667**/**668** of prosthesis **602** so as to partially define openings **632** and **660**. This allows sutures to be threaded through openings **632** and **660**, as desired, to secure soft tissue and filamentary fixation device **640** to prosthesis **602**.

(75) Furthermore, filamentary fixation device **640** may arrive to the operating theater pre-loaded to prosthesis **602** so that its free ends **643** extend in a posterior to anterior direction through openings **658**. The operator may optionally thread a suture through device **640** and through opening **660** in order to further secure device **640** to prosthesis. Alternatively, such suture may be pre-threaded to device **640** and prosthesis **602** before arriving to the operating theater.

(76) In a method for attaching the soft tissue to joint replacement prosthesis **602**, a similar method as shown in FIG. **8** is used with the joint replacement assembly **600** depicted in FIGS. **14**A-**14**E. In this regard, a patellar tendon **902** is detached from the tibial tubercle and a proximal section of the tibia is resected at a location along the tibial shaft so that the removed bone includes the tibial tubercle. In addition, prosthesis **602** is assembled by connecting sleeve **662** thereto. However, sleeve **662** may be pre-assembled prior to delivery to the operating theater. Thereafter, filamentary fixation device **640** is engaged to joint prosthesis **602** by inserting ends **643** of filamentary fixation device **640** into openings **658** so that inner surface **640***b* is brought into communication with a posterior surface **634***a* of metaphyseal portion **634**. Prosthesis **602** is then connected to the bone via a connection portion **604** and/or intramedullary stem (not shown) at the distal end of diaphyseal portion **604**.
(77) Once assembly **600** is mounted to the tibia, soft tissue is secured to joint replacement prosthesis **602** via filamentary device **640**. In order to re-secure the detached patellar tendon **902**, the patellar tendon **902** is positioned adjacent tab portion **663** of sleeve **662** so that tendon **902** contacts the porous structure thereof. Free ends **643** of filamentary device **640** are then wrapped tightly about the tendon **902** (see e.g., FIG. **8**) and sewn thereto such that tendon **902** is sandwiched between porous sleeve **662** and filamentary fixation device **640**. This allows tissue to grow from tendon **902** into the porous sleeve **662** and the porous, mesh-like structure of filamentary device **640**. Moreover, a muscle, such as muscle **904** in FIG. **8** may be sewn to the fixation device **640** at an anterior side of prosthesis **602**. In addition, a suture or cerclage wire may be threaded through openings **632** and/or **660** and passed through tendon **902** and/or muscle **904**. Thus, this configuration allows for a traditional connection to implant **602** via openings **632** and **660** and also allows for the soft tissue **902**, **904** to grow into filamentary component **640** thereby providing a strong initial and long-term connection. Thus, filamentary component **640** provides a soft tissue ingrowth structure to prosthesis **602**.
(78) Joint replacement assembly **700**, as shown in FIGS. **15**A-**15**C, is yet another embodiment of the disclosure and generally includes a joint prosthesis **702**, cannulated screws, **760** and a connection plate or ingrowth plate **750**. Joint prosthesis **702** is similar to prosthesis **302** in that it includes a metaphyseal portion **734** configured for connection to another joint prosthesis, such as a femoral component, and a diaphyseal portion **705** with a connection feature **704** that allows prosthesis **702** to be connected to a resected portion of a tibial shaft.
(79) Metaphyseal portion **734** is connected to a proximal end of diaphyseal portion **705** so that prosthesis **702** tapers outwardly in a distal to proximal direction. Metaphyseal portion **734** includes one or more openings **754** that extend entirely through prosthesis **702** in an anteroposterior direction and that are each configured to receive a cannulated screw **760**. Metaphyseal portion **734** also includes a recessed portion **731** that matches the shape and size of plate **750**, as described in more detail below.
(80) Cannulated screws **760** have a shaft **761**, a head **763**, and a through-opening **762** extending through the length of screw **760**. The shaft **761** is generally cylindrical and comprises the majority of the screw **760**. The distal end of screw **760** includes a threaded portion **765**. Screw **760** is cannulated so that sutures **706** can be threaded through the length of screw **760**. This allows screws **760** to be inserted into openings **754** so that through-opening **762** of each screw **760** provides a passage for a suture through prosthesis **702**. The threaded portion **765** of screw **760** is configured to threadedly engage threaded openings **752** in plate so as to secure plate **750** to prosthesis **702**, as described below.
(81) In the embodiment shown, plate **750** has a “T” like shape, in which plate **750** has a body **751** and two projections or wings **755** extending from body **751** in a lateral-medial direction. Plate **750** further includes threaded openings **752** that extend through the thickness of plate **750** and correspond to the location of openings **754** in prosthesis **702** such that when plate **750** is inserted into recess **731**, openings **752** and **754** align. Openings **752** are threaded to so that screws **760** can

secure plate **750** to prosthesis **702**. Plate **750** is preferably composed of a porous material, such as titanium foam, that supports tissue in-growth. However, openings **752** may have a solid metallic structure to provide sufficient strength for a threaded connection with screws **760**. As mentioned previously, plate **750** has openings **752** with threads to receive screw **760**. As seen in FIG. **17**, openings **752** appear to be where the body **751** of plate **750** meets projections **755**. However, openings **752** can be located anywhere on plate **750** so long as openings **754** on body **730** correspond. However, it is preferable that openings **754** and openings **752** are located at respective lateral and medial sides of a longitudinal axis of prosthesis **702** so that a suture or the like can be passed from one side of axis to another to secure soft tissue to prosthesis **702**. Moreover, plate **750** does not have to be limited to a "T" shape. For example, plate could just be rectangular and not include projections **755**. However, in such embodiment, plate **750** would have a larger lateral-medial width than the plate depicted. Other exemplary configurations of plates are describe in more detail below.

(82) In a method of using assembly **700** to connect a patellar tendon thereto, plate **750** may be connected to prosthesis **702** via cannulated screws **760**. This may be done in the operating theater or prior to delivery to the operating theater. Prosthesis **702** may then be connected to a proximal end of a resected tibia and the patellar tendon positioned adjacent porous plate **750**. Suture or wire may then be threaded through cannulated screws **760** about metaphyseal portion **734** and through and/or about the patellar tendon so as to secure the patellar tendon against porous plate **750** so that tissue from the tendon can grow into its porous structure. Thus, while assembly **700** may not include a filamentary fixation device that would allow for tissue growth therein like that of assembly **600**, assembly **700** still allows for tissue in-growth into the porous plate **750** at the anterior face of prosthesis **702**. Thus, a filamentary device, such as strand of suture or wire, positioned through cannulated screws **760** can help provide immediate fixation, while porous plate **750** facilitates long-term fixation.

(83) However, it is contemplated that assembly **700** can also include a filamentary fixation device to further enhance long term fixation. For example, filamentary fixation device **340** may be slid over diaphyseal portion **705** so that portion **705** is received within opening **346**. The patellar tendon may sutured to second portion **342** as described above with respect to FIG. **8**. In this regard, the patella **902** may be sandwiched between porous plate **750** and second portion **342** of device **340** so that the soft tissue may grow into bone plate **750** and second portion **342**.

(84) Joint replacement assembly **800**, as shown in FIG. **16**, depicts a further joint replacement assembly embodiment of the present disclosure. Joint replacement assembly **800** is similar to assemblies **600** and **700**. In this regard, assembly **800** includes a joint prosthesis **802**, cannulated screws **862**, and a connection plate or ingrowth plate **850** that is of a porous material or has a porous exterior surface connectable to the joint prosthesis **802** similar to that of assembly **700**. However, unlike assembly **700**, four cannulated screws **860** are used to connect plate **850** to prosthesis **802**. Thus, it should be understood that two or more cannulated screws **860** may be used to connect a porous plate to an underlying prosthesis. Also, unlike in assembly **700** but similar to assembly **600**, prosthesis **802** includes longitudinal slots similar to that of slots **658**. This allows filamentary fixation device **640** to be used in assembly **800** to help secure soft tissue thereto as described above. Also, as shown, plate **850** has a rough hour-glass shape which helps occupy as much space as possible between longitudinal slots or through-slots **858** to help ensure contact with soft tissue for growth therein.

(85) Soft tissue is connected to prosthesis **802** similar to that of prosthesis **602** and **702**. In this regard, a patellar tendon that has been disconnected from a tibia is positioned against porous plate **850**. Filamentary fixation device **640** is threaded through slots **858** and free ends **643** thereof are placed over the tendon and sewn together via suture or wire, as described with respect to assembly **600**. In addition, the tendon may be further secured via sutures or wire through cannulated screws **860**, as described with respect to assembly **700**. Thus, this configuration allows for a connection to

implant **802** via the cannulation in screws **860** to provide for a strong immediate connection and also allows for the soft tissue, such as tissue **902**, **904** of FIG. **8**, to grow into filamentary fixation device **640** and porous plate **850** thereby providing a strong long-term connection.

(86) FIG. **17** depicts another joint replacement assembly **900** similar to that of **800** and is accorded like reference numerals, but within the **900** series of numbers. However, the difference between assembly **800** and **900** is that assembly **900** includes a connection plate or ingrowth plate **950** that is of a porous material or has a porous exterior surface with a depression **952** on its anterior or outer surface. This depression **952** allows a portion of bone, such as a resected tibial tubercle with attached tendon, to be received therein and so that such bone can grow into the porous structure of plate **950**. Depression **952** provides a relief for the resected bone with tendon attached thereto to seat at a natural anatomic height relative to the anterior face of prosthesis **902**. Thus, the method is identical to that of assembly **800** with the difference being that bone underlying the patella is excised and connected to plate within its anterior depression **952**.

(87) FIGS. **18**A-**18**E depict another joint replacement assembly **1000** similar to that of assemblies **800** and **900**. In this regard, assembly **1000** includes a joint prosthesis **1002** that includes a metaphyseal portion **1034** and a diaphyseal portion **1005** and is generally constructed as a tibial prosthesis. Moreover, metaphyseal portion **1034** includes a tray region **1042** for a tibial insert, a plurality of through-holes **1054**, **1056** configured to each receive a cannulated screw, such as screws **860** and **960**, an anterior recessed portion **1031** for receipt of an ingrowth plate, such as plates **850** and **950**, and through-slots **1058** for receipt of a filamentary fixation device, such as filamentary device **640**.

(88) However, through-slots **1058** are differently configured than slots **858** and **958**. In this regard, slots **1058** each extend through posterior and anterior sides of prosthesis **1002** such that each slot **1058** defines an anterior or first aperture **1057***a* and posterior or second aperture **1057***b*. The posterior aperture **1057***b*, as best shown in FIG. **18**C, is pill-shaped such that it defines a longitudinal axis **1058***b* that is parallel to a central axis of prosthesis **1002**. The anterior aperture **1057***a*, as best shown in FIG. **18**A, is triangular, as illustrated by the superimposed right triangle **1058***a*. However, the apices of the triangle formed by anterior aperture **1057***a* are rounded rather than pointed. In addition, each slot **1058** is defined along its anteroposterior traversal by first and second opposing sidewalls **1051***a-b*. The first sidewall **1051***a* has a constant superior-inferior height along the traversal of each slot **1058** from posterior aperture **1057***b* to anterior aperture **1057***a*. However, the second sidewall **1051***b* gradually shortens in height along the traversal of each slot **1058** from posterior aperture **1057***b* to anterior aperture **1057***b*. In addition, first wall **1051***a* is generally parallel to a central axis **1004** of prosthesis at both the anterior and posterior aperture **1057***a-b* and also therebetween. However, second wall **1051***b*, while being substantially parallel to central axis **1004** at the posterior aperture **1057***b*, gradually tilts away from central axis **1004** as second wall **1051***b* extends posterior to anterior so that second wall **1051***b* adjacent the anterior aperture **1057***a* is canted away from axis **1004** and forms the hypotenuse of the triangular-shaped anterior aperture **1057***a*. Thus, a plane tangent to second sidewall **1051***b* intersects central axis **1004**. Moreover, in the particular embodiment depicted, second sidewall **1051***b* also curves inwardly about central axis **1004** in a posterior to anterior direction, as illustrated by the arc line overlay **1053** in FIG. **18**B. In other words, second sidewall **1051***b* follows a curved path so that second sidewall **1051***b* itself defines a curved surface. However, it is contemplated that second sidewall **1051***b* may be planar while having other characteristics described above, such as being canted away from central axis **1004**. It is also contemplated that both first and second apertures **1057***a-b* may be triangular.

(89) The depicted embodiment allows filamentary fixation device **640** to be threaded through slots **1058** similar to that described above with respect to assemblies **800** and **900** so that a segment **641** of device **640** that spans between slots **1058** conforms to the posterior side of prosthesis **1002** and is pressed flat against the posterior side of prosthesis **1002**, as best shown in FIG. **18**D. In this

regard, the flat configuration of filamentary device **640** and the orientation of slots **1058** so that they form a posterior aperture **1057***b* that is substantially parallel to the central axis **1004** helps provide a low profile of device **640** at the posterior side of prosthesis **1002** so as to minimize soft tissue irritation while also distributing loads over a broad surface area for durability. In addition, each free end **643** of device **640**, as they follow second wall **1051***b* of their respective apertures **1058**, turns so that their axes are no longer horizontal, but instead are angled superiorly so that free ends **643** cross in an X-shaped arrangement, as best shown in FIG. **18**E. This arrangement helps redirect free ends **643** so that they do not interfere with each other as they extend over the anterior side of prosthesis **1002** and also help distribute loads applied to filamentary fixation **640** device to walls **1051***b*. In other words, a patella sewn to filamentary fixation device **640** will have a tendency to be pulled superiorly by the extensor mechanism during normal movement of the knee joint. Thus, loads applied to device **640** via the patella will tend to pull on free ends **640** superiorly. With assembly **1000**, when such loads are applied to device **640**, free ends **640** will be further compressed to surfaces **1051***s* which further helps distribute the loads over a broad area to maximize durability of device **640**. In addition, the curvature of second surfaces **1051***a* and their angled orientation helps prevent filamentary device **640** from being bunched up at a superior end of each slot **1058**.

(90) Several filamentary fixation devices, such as filamentary fixation devices **120**, **220**, **340**, **500**, and **640**, are described herein. Such fixation devices may be made by folding over and or tubularizing sheets of synthetic mesh, such as Marlex mesh manufactured by CR Bard, Inc., and then shaping such mesh into the desired form. However, folding over or rolling sheets of mesh material can result in a stiff, bulky structure. In addition, sutures threaded through such structures cut through or rip through edges of the mesh material as the only structure preventing this are the woven fibers of the filamentary material which typically loosely engage the suture.

(91) In order to address some of the existing problems with surgical mesh in extensor mechanism repair, filamentary fixation devices **120**, **220**, **340**, **500**, and **640** described herein and other filamentary fixation devices not described herein can be made by layering individual sheets of synthetic mesh/filamentary material and heat sealing/bonding the sheets of material together to form an integrated structure.

(92) An embodiment of such integrated filamentary fixation structure **1100′** is depicted in FIG. **19**C-**19**E. Filamentary fixation device **1100′** is similar to filamentary device **640** and, therefore, can be used in the same manner described above. Filamentary fixation device **1100′** begins as a plurality of layers **1102** of mesh material that are stacked onto one another to form a construct **1100** having a desired width “W”, length “L”, and thickness “T”, as shown in FIGS. **19**A and **19**B. In this regard, the construct **1100** can have any number of layers of material, such as 2 to 20, but preferably 8 to 12 layers. The construct **1100** is then heat sealed through its full thickness so that each layer **1102** is connected to an adjacent layer in order to form device **1100′**. In some embodiments, large sheets of material may be layered, heat sealed, and then cut to the desired dimensions to form device **1100′**. In addition, it is contemplated that other means of connection between layers **1102** is contemplated, such as sonic welding, adhesives, and the like.

(93) As shown in FIG. **19**C-**19**E, filamentary fixation device **1100′** includes a plurality of seams **1104** that are formed so that they each extend continuously along the entire length thereof and through the entire thickness thereof. In this regard, the length of each seam **1104** is in a direction of the expected load. Thus, in some circumstances, where the expected load may be multi-directional, it is contemplated that there may be overlapping seams that extend in different directions, such as along both the length and width of device **1100′**. In the embodiment depicted, a first and second seam **1104***a-b* are formed at the side edges of device **1100′** and a third seam **1104***c* is formed down a centerline of device **1100′**. In this regard, each layer **1102** is free to move relative to adjacent layers between the seams **1104***a-c*, as depicted in the cross-sections of FIGS. **19**D and **19**E. Some embodiments may have more than three continuous seams **1104**, such as 4 to 8 seams. Moreover,

seams **1104** may be positioned relative to each other at consistent intervals or at differing intervals such that the distance between each seam **1104** differs from one seam to the next.

(94) FIG. **19**E illustrates various suture pathways that may be used to connect soft tissue to device **1100**′. As shown, a suture may be threaded through each layer of device **1100**′ between adjacent seams. In this regard, seams act as a structural barrier that helps prevent sutures from cutting entirely through device and helps maintain the sutures in their respective lanes between seams.

(95) FIG. **19**F depicts an alternative fixation device **1100**″. Fixation device **1100**″ has a plurality of interrupted or discontinuous seams **1104**′, whereas device **1100**′ has a plurality of continuous seams **1104**. Thus, while each seam **1104**′ similarly extends along the length of device **1100**″ and through its full thickness, each seam **1104**′ is interrupted at predetermined intervals along its length by free segments **1106** which lie along the axis of the seam **1104**′. The layers **1102** at the free segments **1106** are not bonded together and, therefore, are free to move relative to each other. Such an interrupted configuration can provide further flexibility over that of device **1100**′. It is also contemplated that other devices not shown can have a combination of continuous and interrupted seams. For example, side edges of a filamentary fixation device can be sealed with continuous seams, such as seams **1104***a-b*, while one or more seams between such side edges are sealed with interrupted seams, such as seams **1104**′.

(96) Layered mesh with continuous or interrupted seams **1104**, **1104**′ are advantageous because they allow the overall construction of a filamentary fixation device to have a thinner profile than a typical folded or rolled construction. Moreover, such layered construction provides enhanced flexibility as the overall construct is less bulky. In addition, the interrupted seam **1104**′ may provide even further flexibility due to free segments **1106**, as mentioned above. Even further, layering mesh filamentary material and connecting them via a heat fusion or the like allows the layered construction to be cut and shaped much like a textile material so that the layered construct can take on any of the forms of the herein described filamentary fixation devices.

(97) FIGS. **20**A-**20**F depict further joint replacement assembly **1200** similar to that of assembly **1000**. In this regard, assembly **1200** includes a joint prosthesis **1210** that includes a metaphyseal portion **1234** and a diaphyseal portion **1205** and is generally constructed as a tibial prosthesis. Moreover, metaphyseal portion **1234** includes a tray region **1242** for a tibial insert, a plurality of screw openings **1254**, **1256** configured to each receive a screw **1260**, an anterior recessed portion similar to recessed portion **1031**, a connection plate or ingrowth plate **1290** received within such recessed portion, and through-slots or through-openings **1258** for receipt of a filamentary fixation device, such as filamentary devices **640**, **1100**, **1100**′, and **1100**″, for example. Assembly **1200** also includes through-holes **1232** configured to receive sutures or wires, such as cerclage wires, for example.

(98) However, through-slots **1258** are differently configured than slots **1058**. In this regard, slots **1258** each extend through posterior and anterior sides of prosthesis **1210** such that each slot **1258** defines an anterior or first aperture **1257***a* and posterior or second aperture **1257***b*. The posterior and anterior apertures **1257***a-b*, as best shown in FIGS. **20**A and **20**B, are both elliptical in shape such that their respective major axes (i.e., an ellipses major axis) extend in a generally mediolateral direction. In addition, each slot **1258** is defined along its anteroposterior traversal by first and second opposing sidewalls **1251***a-b*. First sidewall **1251***a*, which is closer to a midline longitudinal axis of prosthesis **1210**, is convexly curved generally about such midline axis. Conversely, the second sidewall **1251***b* is concavely curved but also generally about the midline axis of prosthesis **1210**. In addition, as best shown in FIG. **20**E, opening **1258** has a narrower opening in a proximal-distal direction near the center of opening **1258** while being wider nearer the anterior and posterior apertures **1257***a-b*. In other words, opening **1258** gradually narrows from posterior aperture **1257***b* to a center of opening **1258** and then gradually widens from the center of opening **1258** to anterior aperture **1251***a*.

(99) The depicted embodiment allows filamentary fixation device **640** (or also **1100**, **1100**′, **1100**″)

to be threaded through slots **1258** similar to that described above with respect to assembly **1000**. However, the narrowing and widening of openings **1258** in addition to their elliptical shape causes a flat filamentary fixation device to slightly fold over itself as it extends from posterior aperture **1257***b* to anterior aperture **1257***b* and vice versa. This allows filamentary fixation device **640** to conform to convex inner surface **1251***a* while being manipulable at the anterior side of prosthesis **1210** so that free ends **640***a-b* of fixation device **640** can be oriented at any desired angle without awkward crimping or bunching thereof. In this regard, prosthesis **1210** allows fixation device **640** to achieve any of the orientations previously described and more such that an extensor mechanism can be connected thereto in the manner previously described.

(100) Connection plate or ingrowth plate **1290**, as shown in FIGS. **20**C and **20**D, include a plate body **1291** that has a thickness extending between an anterior face **1292** and a posterior face **1295** thereof. Anterior face **1292** (see FIG. **20**D) preferably includes a porous material extending over at least a portion of its area in order to promote tissue ingrowth therein. Plate **1290** includes a proximal portion **1270**, a distal portion **1280**, and an intermediate member **1294** extending therebetween. Distal portion **1280** includes a first and second boss **1281***a-b* extending posteriorly from the posterior face **1295**. A first and second through-opening **1286***a-b* respectively extend through bosses **1281***a-b* from the anterior side to the posterior side of plate body **1291**. Distal portion **1280** also includes a first and second counterbore **1288***a-b* each of which are respectively coaxial with first and second through-openings **1278***a-b*. Such configuration forms a circular rim **1282** between each counterbore and through-opening pairing. First and second openings **1286***a-b* are each configured to receive a threaded shaft **1262** of screw **1260**. However, openings **1286***a-b* each have a cross-sectional dimension smaller than that of a head **1264** of screw **1260** such that head **1264** is prevented from passing therethrough, but is capable of being disposed within each of counterbores **1288***a-b* as best shown in FIGS. **20**E and **20**F. In this regard, screw **1260** is engaged to distal portion **1280** via the anterior side of plate body **1281** such that threaded shaft **1262** is passed through one of counterbores **1288***a-b* and then through respective through-opening **1286***a-b* so that head **1264** rests against rim **1282** from within counter bore **1288***a* or **1288***b*.

(101) Proximal portion **1270** also includes a first and second boss **1271***a-b* extending posteriorly from posterior face **1295**. However, proximal portion **1270** does not include counterbores. In addition, bosses **1271***a-b* each include a respective side-slot **1276***a-b* that defines a U-shaped rim **1272**. A groove **1274** extends along posterior face **1295** in a mediolateral direction and communicates with side-slots **1276***a-b*. A first and second through-opening **1278***a-b* extend through the anterior face **1292** and posterior face **1295** and each intersect a respective side-slot **1276***a-b*. First and second through-openings **1278***a-b* each have a cross-sectional dimension smaller than that of screw head **1264**. Unlike distal portion, a screw **1260** is engaged to proximal portion **1270** via the posterior side of plate body **1291** such that head **1264** is inserted into groove **1274** and slide along posterior surface **1295** and into one of side slots **1276***a-b*. Once disposed within one of side-slots **1276***a-b*, head **1264** rests on rim **1272** and is positioned between rim **1272** and a respective through-opening **1278***a-b*. In this regard, through-opening **1278***a* or **1278***b* aligns with a tool opening within head **1264** of screw so that a tool/driver can engage screw **1260** which is rotatably positioned within one of side-slots **1276***a-b* despite being positioned on the posterior side of plate body **1291**, as best shown in FIG. **20**D. However, threaded shaft **1262** extends posteriorly from side-slot **1276***a-b* for engagement with one of threaded openings **1254** of prosthesis **1210**. This configuration reduces the overall footprint of through-openings **1278***a-b* relative to anterior surface **1292** of plate body **1291** as compared to that of counterbores **1288***a-b*. As such, more surface area of plate body **1291** can be dedicated to a porous surface for ingrowth particularly in the proximal portion **1270** of plate **1290** which is more likely to be in direct contact with soft tissue than distal portion **1280**.

(102) However, it should be understood that proximal portion **1270** can be configured the same as distal portion **1280** such that proximal portion **1270** includes counterbores, such as counterbores

1288*a-b*, for screw loading via the anterior side of plate body **1291**. Alternatively, distal portion **1280** may be configured the same as proximal portion **1270** such that distal portion **1280** includes side-slots, such as side-slots **1276***a-b*, so that screw **1260** can be loaded from the posterior side of plate **1290**. Also, it should be understood that in some embodiments, plate **1290** may not include bosses **1271***a-b* or **1281***a-b*, and may instead just have extra plate thickness. Moreover, in some embodiments, plate **1290** may not include groove **1274**. However, groove **1274** is helpful to make space for head **1264** of screw **1260**.

(103) Proximal portion **1270** and distal portion **1280** are connected via intermediate member **1294** such that plate body **1291** has an hour-glass shape. In other words, proximal portion **1270** and distal portion **1290** are wider than at intermediate portion **1294**. This configuration creates indentations at the sides of plate body **1291** to make space for openings **1258** of FIGS. **20**A and **20**B. FIGS. **20**E and **20**F illustrate the connection between plate **1290** and prosthesis **1210**. In this regard, plate **1290** is disposed in an anterior recess of prosthesis **1210** with conforms to the shape of plate body **1290**. Porous anterior face **1292** is positioned outward at the anterior side of prosthesis **1210** so as to engage tissue when implanted, as described above with respect to other embodiments of the present disclosure. Screws **1260** pass through distal portion **1280** such that threaded shafts **1262** engage threaded openings **1256** and heads **1264** engage circular rim **1282** from within counterbores **1288***a-b*. Additional screws **1260** are positioned at posterior side of plate **1290** within side-slots **1276***a-b* such that threaded shafts **1262** extend from plate body **1291** and into threaded openings **1254**. In this regard, each head **1264** is sandwiched between rim **1272** and at least a portion of the thickness of plate body **1291**.

(104) Although the invention herein has been described with reference to particular embodiments, it is to be understood that these embodiments are merely illustrative of the principles and applications of the present invention. It is therefore to be understood that numerous modifications may be made to the illustrative embodiments and that other arrangements may be devised without departing from the spirit and scope of the present invention as defined by the appended claims. For example, although embodiments of the invention have generally been described in reference to an orthopedic assembly in a tibia, the principles described herein are equally applicable to bones of other joints.

Claims

1. An orthopedic assembly comprising: a joint prosthesis; and a filamentary component having an annular portion and an elongate portion, the annular portion defining an opening that extends from a first side of the annular portion to a second side of the annular portion opposite the first side, and the elongate portion extending from the first side of the annular portion, wherein the opening is configured to receive a portion of the joint prosthesis therein such that a longitudinal axis of the joint prosthesis passes through the opening and is aligned with a central longitudinal axis of the opening, wherein when the joint prosthesis is received in the opening, a length of the elongate portion along the longitudinal axis of the joint prosthesis is greater than a length of the annular portion extending from the first side to the second side along the longitudinal axis of the joint prosthesis, wherein the length of the elongated portion of the filamentary component extends beyond a proximal end of the joint prosthesis in a direction away from the annular portion, the annular portion being in between the proximal end of the joint prosthesis and a distal end of the joint prosthesis, wherein an inner face of the elongated portion of the filamentary component comprises a soft-tissue-engaging surface, and wherein when the joint prosthesis is received in the opening, the elongated portion of the filamentary component is conically curved around a region of the joint prosthesis such that the orthopedic assembly is configured to sandwich soft tissue between the elongated portion and the joint prosthesis to attach the soft tissue therebetween.

2. The orthopedic assembly of claim 1, wherein the filamentary component is made from a material

including one or more of a synthetic polymer, a bioresorbable fiber, a ceramic, a biological material, and a pharmacological agent such that the filamentary component is configured to attach to soft tissue.

3. The orthopedic assembly of claim 1, wherein the portion of the joint prosthesis is received in the opening such that the filamentary component is restrained from withdrawal from the portion in one direction along the longitudinal axis.

4. The orthopedic assembly of claim 1, wherein the annular portion flares outwardly along its length in a direction from a free end of the filamentary component toward the elongate portion.

5. The orthopedic assembly of claim 4, wherein, when assembled, an inner surface of the annular portion is disposed against a tapered outer surface of the joint prosthesis.

6. The orthopedic assembly of claim 1, wherein the joint prosthesis is either a humeral or a femoral joint prosthesis including a head, a body and a stem such that when the humeral or the femoral joint prosthesis is received in the filamentary component, the filamentary component is restrained from withdrawal from the filamentary component by the head.

7. The orthopedic assembly of claim 1, wherein the joint prosthesis is a proximal tibial prosthesis including a metaphyseal portion and a diaphyseal portion extending from the metaphyseal portion, the metaphyseal portion having an anterior side and a posterior side arranged such that when the proximal tibial prosthesis is received in the filamentary component, the elongate portion extends along the anterior side of the metaphyseal portion.

8. The orthopedic assembly of claim 7, further comprising a suture extending through a suture hole of the metaphyseal portion and through the elongated portion of the filamentary component, wherein the elongated portion of the filamentary component is disposed adjacent to the suture hole extending through the metaphyseal portion.

9. An orthopedic system comprising: a first joint prosthesis including suture holes and having a first end configured to engage with a bone and a second end opposite the first end, the second end configured to interface with a second joint prosthesis; and a filamentary component having a first portion and a second portion, the first portion defining a sleeve having a first length extending from a first side of the sleeve to a second side of the sleeve opposite the first side along a longitudinal axis of the first joint prosthesis and configured to receive the first joint prosthesis, and the second portion having a second length extending from the first side of the sleeve along the longitudinal axis, the second portion being configured to attach to soft tissue, wherein the second length is greater than the first length, wherein the second portion extends beyond a proximal end of the first joint prosthesis in a direction away from the first portion, and the first portion is positioned between the proximal end and a distal end of the first joint prosthesis, wherein an inner face of the second portion of the filamentary component comprises a soft-tissue-engaging surface, and wherein the orthopedic system is configured to receive soft tissue between the filamentary component and the joint prosthesis such that the soft tissue is disposed adjacent to the suture holes of the joint prosthesis.

10. The orthopedic assembly of claim 9, wherein the filamentary component includes a plurality of layers of a mesh material.

11. The orthopedic assembly of claim 10, wherein each layer of the plurality of layers is connected to an adjacent layer by at least one seam formed by heat.

12. The orthopedic assembly of claim 9, wherein the first joint prosthesis is a femur prosthesis that includes a stem and a head, the stem extending from the head and being configured to be inserted into an intramedullary canal of a femur, and the head having a diameter larger than that of the sleeve.

13. The orthopedic assembly of claim 12, wherein the sleeve tapers outwardly in a manner that corresponds with a conically tapered body portion of the femur prosthesis in between the stem and the head such that when assembled, the sleeve bears against the body portion.

14. The orthopedic assembly of claim 9, wherein the second length of the second portion of the filamentary component is greater than a diameter of the sleeve at the first side.